91262 Practice Excellence Questions

Answer Schedule

Q	Answer
1	Maximum volume = 500 cm^3 ($x = 10$ cm, $h = 5$ cm)
2	$r = \dfrac{6}{4+\pi} \approx 0.840$ m; maximum area = $\dfrac{18}{4+\pi} \approx 2.52$ m^2
3	Maximum area = 32 units^2 (vertices at $x = \pm 2$, height 8)
4	Maximum area = 12 cm^2 (4 cm by 3 cm)
5	Width 30 m, length 60 m; maximum area = 1800 m^2
6	Minimum value = 20 (at $x = 2$, $y = 4$)
7	$r = \sqrt[3]{\dfrac{250}{\pi}} \approx 4.30$ cm
8	4 m (4 m from O in the positive direction)
9	Approximately 250 m from the traffic lights (≈ 249.6 m); speed ≈ 31.6 m s^{-1}
10	Approximately 29.8 m s^{-1}
11	(i) No: the car needs 82.5 m to stop (ii) $\dfrac{125}{24} \approx 5.21$ m s^{-2}
12	$c = -7$ or $c = \dfrac{67}{27}$
13	$y = 2x - 4$ and $y = -6x + 4$
14	P$(5, 52)$
15	$a = \dfrac{7}{4}$, $b = -6$, $c = 3$, $d = 0$; $(2, -4)$ is a local minimum
16	$p = 0$, $q = -3$; other stationary point $(-1, 6)$ is a local maximum
17	$y = x^3 - 2x^2 - 4x + 5$; other turning point $\left(-\dfrac{2}{3}, \dfrac{175}{27}\right)$
18	$-6 \le k \le 6$
19	18π cm^3 per cm (at $r = 3$ cm)
20	(i) $-k < x < k$ (ii) $k = \sqrt[3]{2}$
21	$f'(x) = 3x^2 - 2$
22	Normal: $y = \dfrac{1}{2}x - \dfrac{7}{2}$; meets the curve again at $\left(\dfrac{7}{2}, -\dfrac{7}{4}\right)$
23	18 units^2 (tangent $y = 12 - 4x$)
24	Smallest gradient $= -3$, at the point $(2, 3)$
25	(i) $t = 2.5$ s; 12.5 m apart (ii) $t = 5$ s
26	P$(2, 4)$; shortest distance = $\dfrac{7\sqrt{17}}{2} \approx 14.4$
27	15 items per day; maximum profit = \$1350
28	$r = 4$ cm; maximum volume = $64\pi \approx 201$ cm^3
29	$y = x^3 - 2x^2 + x + 2$

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30	$f(x)$: local maximum $(-1, 4)$, local minimum $(3, -3)$, passing through $(0, 2)$. $f'(x)$: crosses the x-axis at $x = -1$ and $x = 3$, positive outside this interval and negative between
31	$\dfrac{dy}{dx} = 3x^2 + 3 \geq 3$ for all x; tangent with smallest gradient: $y = 3x$
32	Shown: tangents meet at $\left(\dfrac{a+b}{2},\ ab\right)$, and perpendicular gives $4ab = -1$, so $y = -\dfrac{1}{4}$
33	2.5 cm^2 per cm
34	$x = 7.5$ cm (maximum area 112.5 cm^2)
35	$-4 < k < 0$
36	$\left(\sqrt{3},\ 2\sqrt{3}\right)$
37	5.5 seconds
38	$f(x) = -\dfrac{1}{4}(x-3)^2 + 2$ $\left(= -\dfrac{1}{4}x^2 + \dfrac{3}{2}x - \dfrac{1}{4}\right)$
39	$v = 10\ \text{km h}^{-1}$; minimum cost = \$1800
40	(i) 16 m (at $t = 2$ s) (ii) $t = 2\sqrt{3} \approx 3.46$ s